
Comparison of a newly introduced pressure insole to existing foot pressure measurement technologies

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A summary document comparing a newly introduced pressure measurement insole with an established leading competitor. An additional comparison between these two pressure measurement technologies and a standardized force plat is also provided.

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Introduction

Orpyx has been developing pressure insoles for medical purposes for the last few years, with a focus on monitoring and assisting patients with diabetes who experience reduced foot sensitivity. Their current pressure insole design (Orpyx LogR) has eight embedded piezo-resistive strain gauge pressure sensors and are designed to send direct feedback to the patient when high pressure is maintained around a specific location over a long period of time. Orpyx is now developing a new line of pressure sensitive insoles with the goal of reaching further markets, such as the sports, orthopedics and biomechanics communities. Therefore, Orpyx has developed the Orpyx LogR Gen 2, which consists of 37 piezo-resistive strain gauge pressure sensors and includes a dorsum initial measurement unit (IMU) that can be mounted to the shoelaces. In order to collect general advice regarding the development of this new prototype and to assess its performance with respect to other leading competitors, Biomechanigg Sport & Health Research Inc. was asked to conduct a study with the objective to:

Compare the new Orpyx LogR Gen 2 prototype to the well-established Pedar System (Novel, Germany) and a Kistler force plate (Kistler Group, Switzerland).

In order to address this objective, four different studies were performed:

1. Estimations of the centre of pressure (COP) trajectory using Pedar data for different movements and different sensors configurations;
2. Evaluation of the appropriate insole placement by comparing pressure distribution patterns collected with (a) Pedar only and (b) Pedar placed on top of the Orpyx LogR;
3. Comparison between the Orpyx LogR Gen 2 prototype and the Pedar insoles;
4. Comparisons between the Orpyx LogR Gen 2 prototype, the Pedar insoles and the Kistler force.

Methods

Recruitment

For studies 1 and 2, Pedar data were collected on a single subject (female, age: 23 shoe size: US 7). For study 3, Pedar and Orpyx data were collected on one subject (male, age: 32, shoe size: US 10). For study 4, data were collected on one subject using the Orpyx insoles and the Kistler force plate (male, age: 27, shoes size: US 10).

Insoles details

The different characteristics of each insole (Pedar and Orpyx LogR Gen2) are presented in Table 1 and Figure 1. The technical characteristics of the IMU embedded in the Orpyx LogR Gen 2 system are presented in Table 2, as well as its position on the corresponding shoe pod in Figure 2.

Table 1 Technical information of the Pedar system and Orpyx LogR Gen 2 insoles

	Pedar	Orpyx LogR Gen 2
Number of sensors	99	37
Sampling frequency [Hz]	Part 1 and 2: 200 Part 3 and 4: 100	256
Insole length [cm]	Black: 27.25	27.76
Insole width [cm]	Black: 8.7	~ 9
Max loading per cell [kPa]	600	600
Pressure Sensors	capacitive	piezo-resistive strain gauge

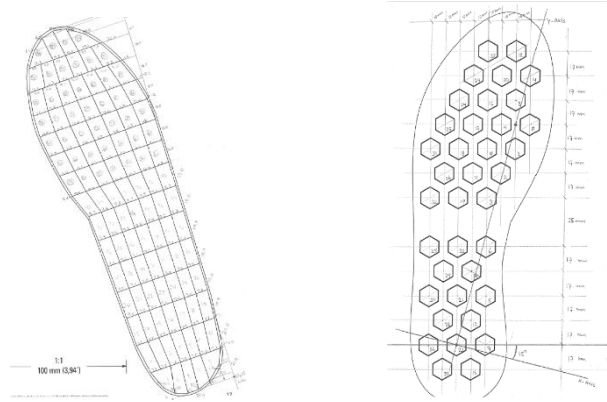


Figure 1 Pressure sensor distribution of the Pedar system (left) and the Orpyx LogR Gen 2 insole (right)

Table 2 Technical information of the IMU sensor embedded in the Orpyx Log R Gen 2 system

	Rate	Sampling Frequency
Gyroscope	2000 °/s	256 Hz
Accelerometer	16 g	256 Hz
Magnetometer	±4900 μT	80 Hz



Figure 2 Position of the IMU sensor (red square) on the shoe pod

Testing Protocol

For studies 1-3, subjects were asked to perform two standing, walking, and running trials of 30-60 seconds each, along with three drop jumps from a height of 33 cm and four side shuffles (two to the left and two to the right). The running and walking trials took place on a treadmill at a self-selected speed. During study 1, data were only collected for the right foot with the Pedar system. During study 2, both insoles were placed in the shoe (Pedar on top of Orpyx LogR), but data were only collected with the Pedar system for the right foot. During study 3, data were collected with both systems simultaneously (Pedar on top of Orpyx LogR Gen 2) for both feet, but only the left insole was used for data analysis. All subjects used their own running shoes during the testing session. Study 4 involved seven barefoot walking trials over the left insole of both systems, which were positioned and fixed on top of the force plate.

Data processing

In study 1, the data from the 99 sensors of the Pedar system were interpolated to generate a comparable configuration of sensors to the two Orpyx designs (Figure 3), or artificial configurations with fewer sensors (Figure 4). The impact of these different configurations on the COP trajectory was evaluated. A threshold of 30N was used for the step detection, and data were not normalized.

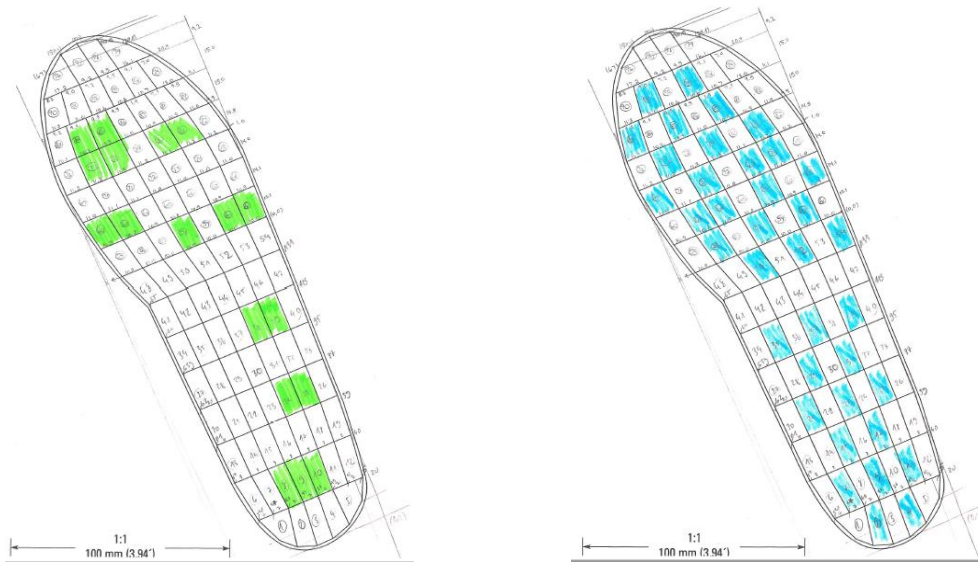


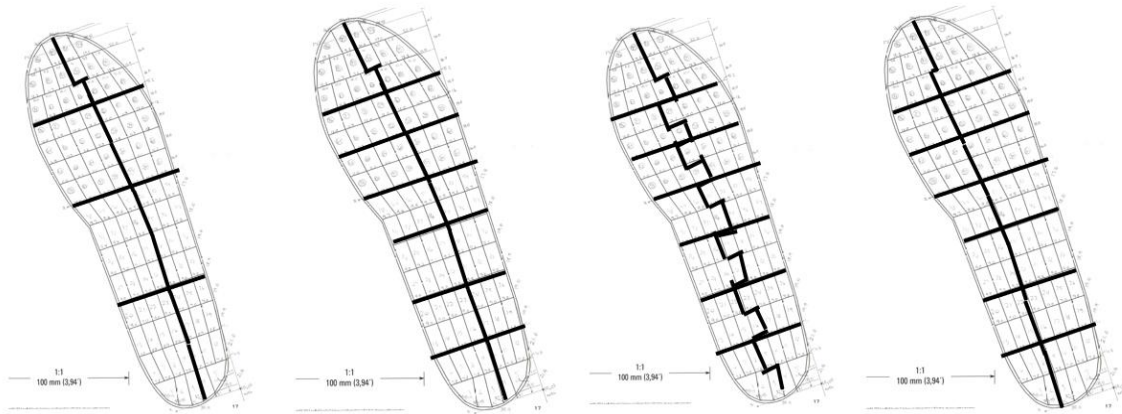
Figure 3 Artificial sensor configurations generated to replicate the configurations of the Orpyx LogR (green, left) and the Orpyx LogR Gen 2 (blue, right).

8 Sensors

14 Sensors medial

14 Sensors equal

14 Sensors lateral



12 Sensors

22 Sensors

32 Sensors

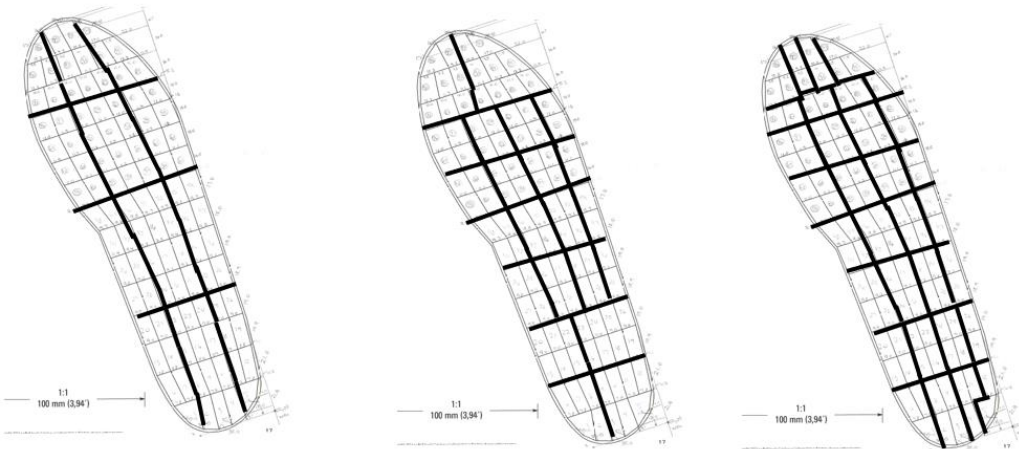


Figure 4 Artificially generated sensor configurations created using data from the 99 Pedar sensors

In study 2, the different pressure distribution patterns resulting from the different configurations (Pedar alone vs. Pedar on top of Orpyx LogR) were compared to analyze the impact of a second insole on the overall pressure pattern.

In study 3, each trial was normalized to 100 frames to facilitate the processing of the corresponding data. Due to technical issues with the force values obtained using the Orpyx LogR Gen 2 insoles, a higher threshold of 50N was used for the step detection. The mean total excursion was calculated as the mean difference between the COP trajectories of both systems between 5% and 95% of the stance phase. The residual

was defined as the sum of the distances between the COP of both systems between 5% and 95% of the stance phase.

In study 4, the two insole systems were compared with a standardized force plate with respect to ground reaction force using the difference in peak force and the root-mean-square error (RMSE). For the Orpyx LogR Gen 2 insole, the force located on the areas of the insole without pressure sensors were estimated based on measured pressure values.

Variables of Interest

The focus was put on the comparison of the COP trajectory and the vertical ground reaction force (GRF) between the two pressure insole systems. The variables analyzed in each study are listed below.

Study 1	COP trajectory
Study 2	Pressure distribution patterns
Study 3	COP trajectory Medial-lateral COP excursion Anterior-posterior COP excursion Total COP excursion Step to step variability Total force
Study 4	Total force

Results

Study 1- Estimation of COP trajectory using different sensor configurations

The results of the two configurations generated to replicate the Orpyx LogR and LogR Gen 2 sensors distributions are shown in Figure 5. The results show some very slight differences between the two artificial configurations and the original setting that covers the entire insole.

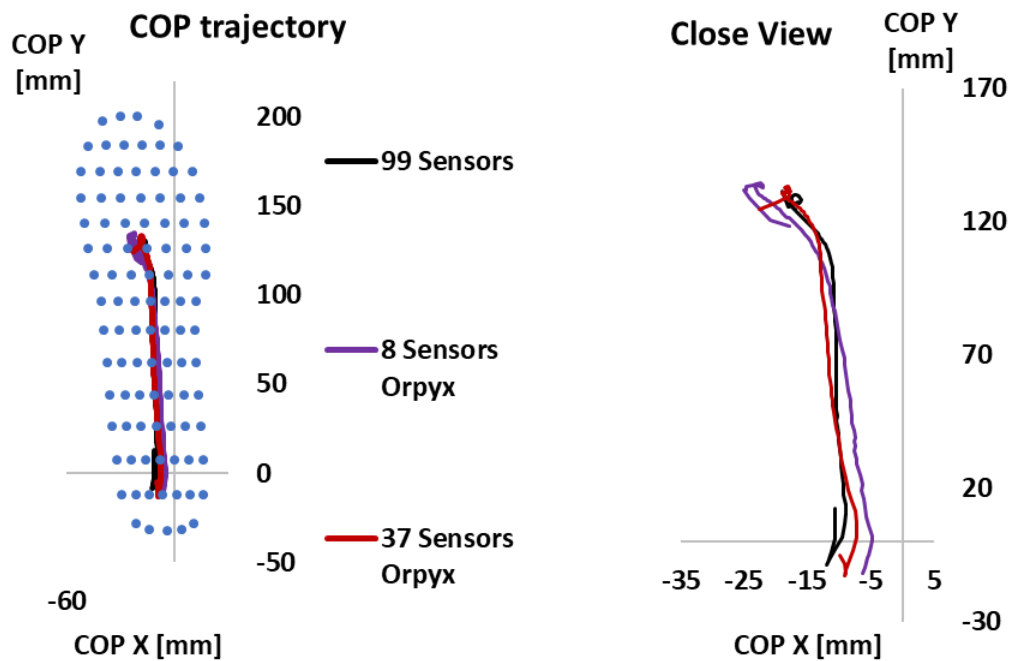


Figure 5 Walking COP trajectories comparing the original 99 sensors to the artificially generated 8 and 37 sensors configurations replicated the settings of the Orpyx LogR and LogR Gen 2 insoles.

Figure 6 shows the differences between the COP trajectories calculated for different sensor configurations during walking. Other movements are presented in Appendix A.

The results also show a high similarity between the original 99-sensor setting and each of the artificially generated configurations. In Figure 7, different versions of the 14-sensor configuration were tested (medial, central, and lateral). The results show a slightly higher accuracy for the central distribution compared to the other two configurations.

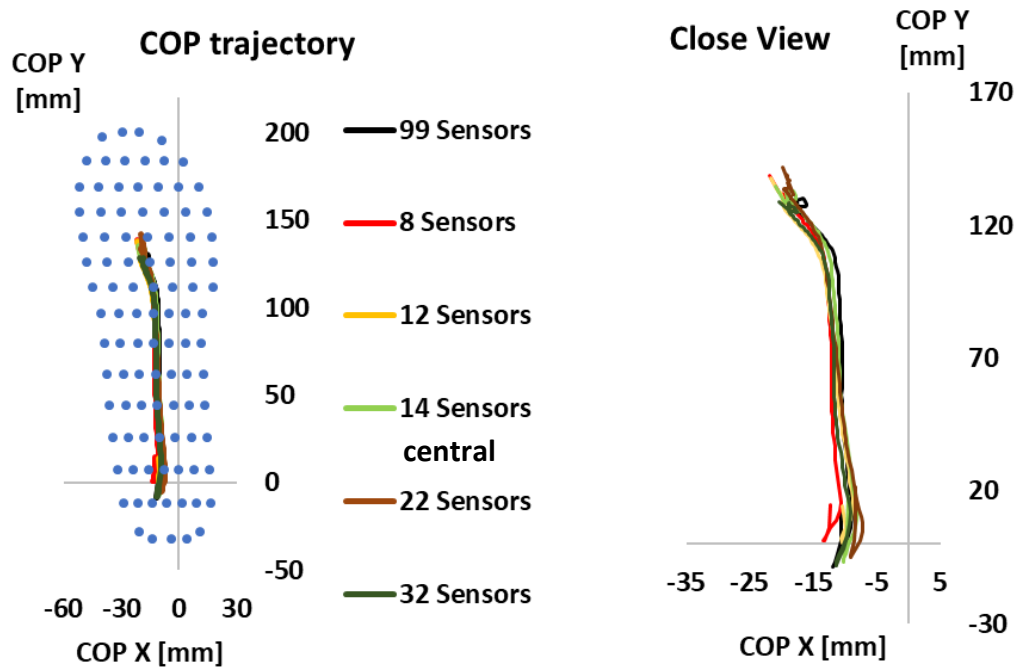


Figure 6 Walking COP trajectories comparing the original 99 sensors to the artificially generated 8, 12, 14 central, 22 and 32 sensors configurations.

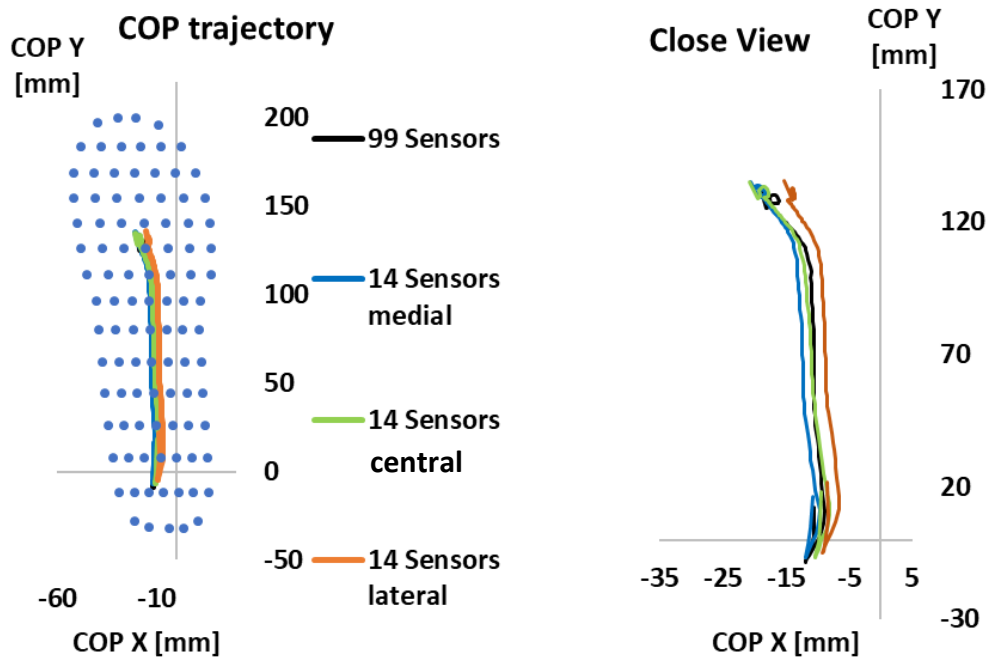


Figure 7 Walking COP trajectories comparing the original 99 sensors to the artificially generated 14 medial, 14 equal and 14 lateral sensors configurations.

Study 2- Impact of insole placement on pressure distribution patterns

Figure 8 shows a comparison of the plantar pressure distribution patterns calculated during two steps of the same walking trial when only the Pedar insoles were placed in the subject's shoes. Figure 9 shows a similar comparison when the Pedar insoles were placed on top of the Orpyx LogR insoles. These two figures show that the variability between steps (highlighted by the difference plots between the left and middle images) is much higher than the differences observed when comparing the Pedar insole to the Pedar insole positioned on top of the Orpyx LogR insole. More specifically, the first step observed in Figures 8 and 9 (left images) are comparable in terms of pressure pattern (same with the two middle images). This demonstrates that the impact of placing the Pedar system on top of the Orpyx LogR insoles only changes the amount of pressure, but not the pressure pattern when visually comparing them. Therefore, it was decided that both insoles would be placed in the subject's shoes during the following study, which aimed to compare the two systems during different tasks.

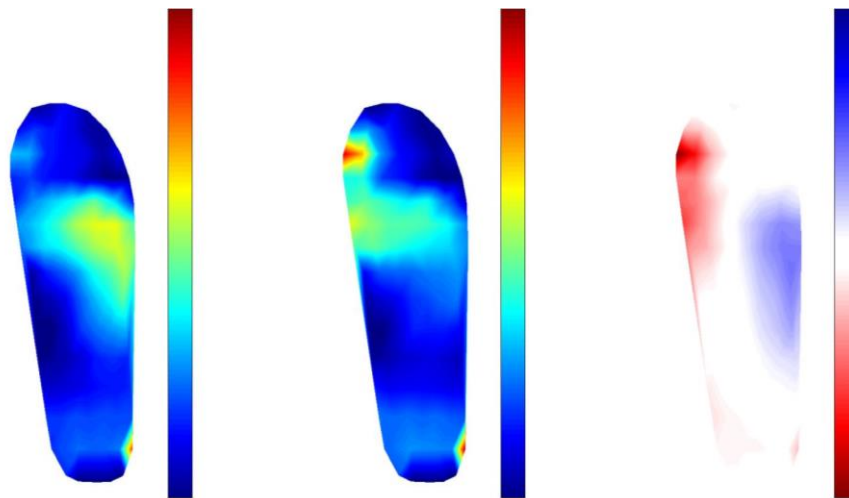


Figure 8 Pressure distribution patterns calculated using the Pedar insoles only during two separate steps (left and middle images) of the same walking trial. The last image (right) represents the difference between the two compared steps.

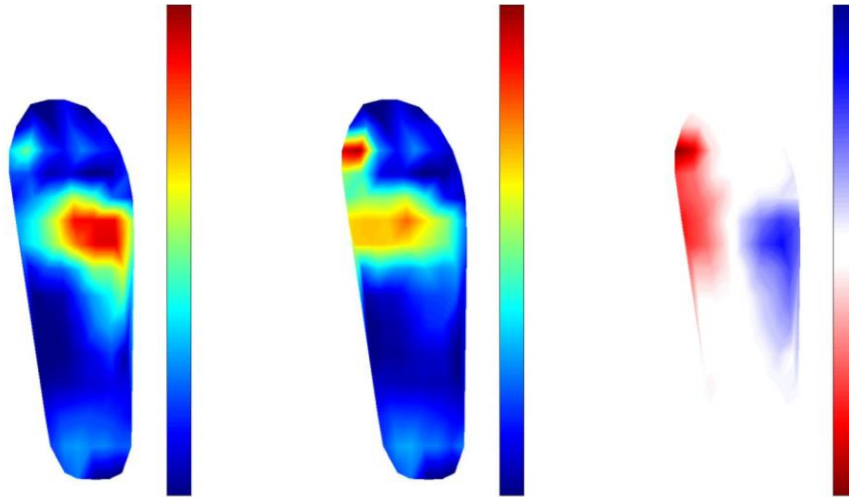


Figure 9 Pressure distribution patterns calculated using the Pedar insoles placed on top of the Orpyx LogR Gen 2 insoles during two separate steps (left and middle images) of the same walking trial. The last image (right) represents the difference between the two compared steps.

Study 3- Comparison between the Orpyx LogR Gen 2 pressure insoles and the Pedar system

COP trajectory and excursion

In this section, the results comparing the COP trajectories of the Orpyx LogR Gen 2 with the Pedar system are presented during walking (Figure 10), running (Figure 13), side shuffle (Figure 16), drop jump (Figure 17) and standing (Figure 18). For the walking and running trials, the antero-posterior (AP) and medio-lateral (ML) excursions were calculated from 0% to 100% of the stance phase (Figures 11-12 and 14-15, respectively).

During most tasks, a slight lateral shift of the COP trajectory was observed with the Orpyx LogR Gen 2 compared to the Pedar insoles. In addition, the most complex movements (i.e. side shuffle, drop jump) resulted in larger variations of the COP trajectory. In terms of excursion during walking and running, it was mostly observed in the AP direction.

Walking:

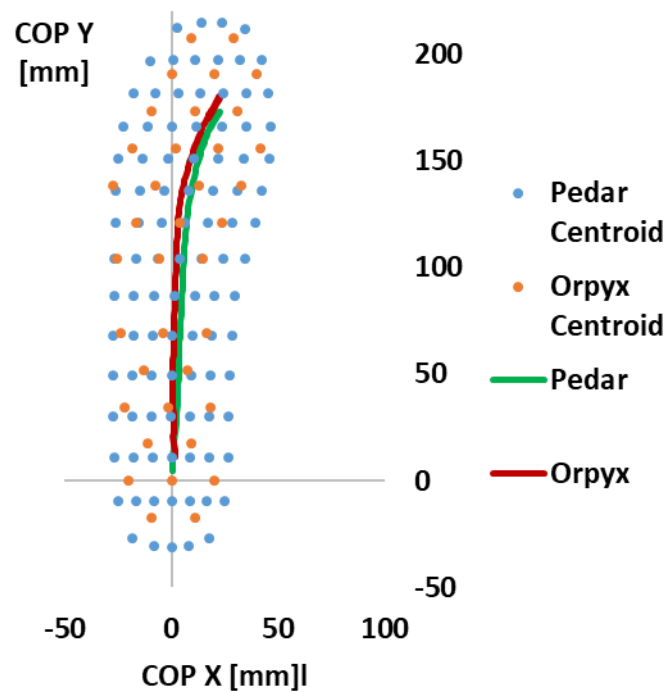


Figure 10 Mean COP trajectories calculated with the Pedar (green) and Orpyx LogR Gen 2 (red) insoles during walking (n = 60 steps).

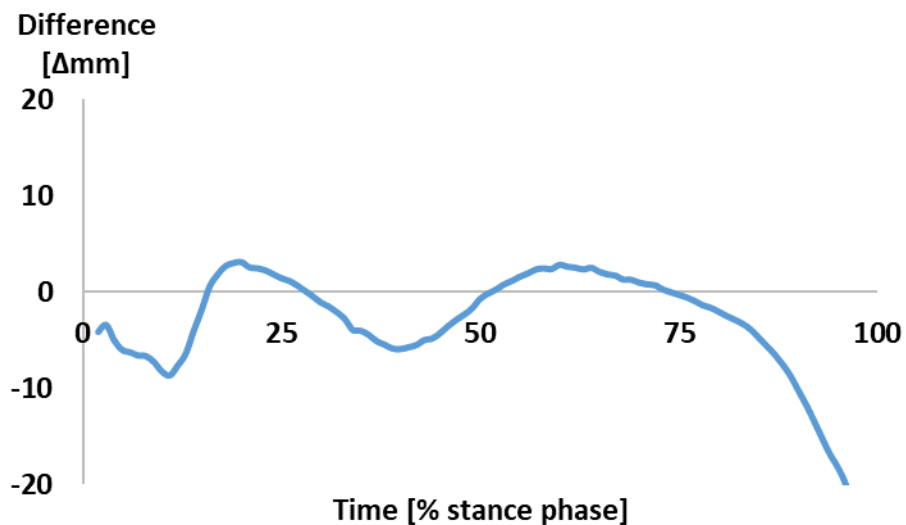


Figure 11 Difference between the mean AP excursion in the Pedar and the Orpyx LogR Gen 2 during a walking trial. A positive value indicates a more posterior COP in the Orpyx LogR Gen 2 insoles, and a negative value corresponds to a more anterior COP in the Orpyx LogR Gen 2 insoles.

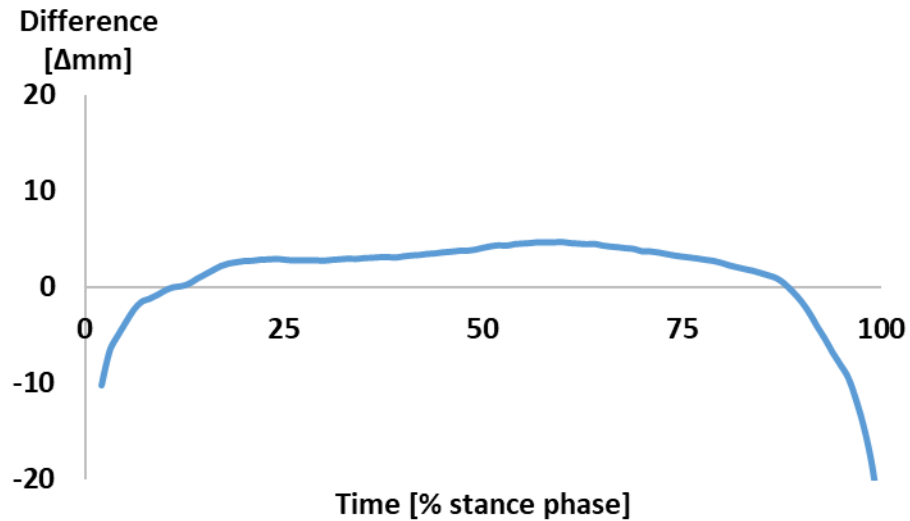


Figure 12 Difference between the mean ML excursion in the Pedar and the Orpyx LogR Gen 2 during a walking trial. A positive value indicates a more lateral COP in the Orpyx LogR Gen 2 insoles, and a negative value corresponds to a more medial COP in the Orpyx LogR Gen 2 insoles.

Running:

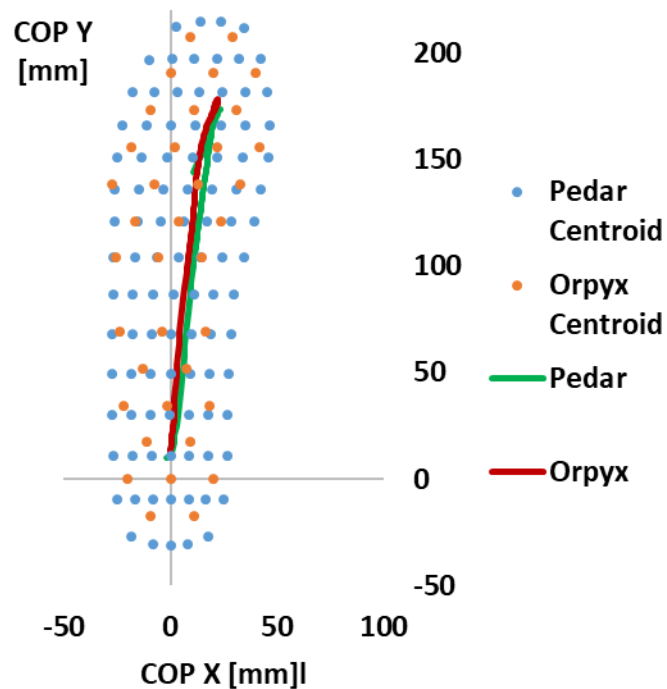


Figure 13 Mean COP trajectories calculated with the Pedar (green) and Orpyx LogR Gen 2 (red) insoles during a running trial (n= 96 steps).

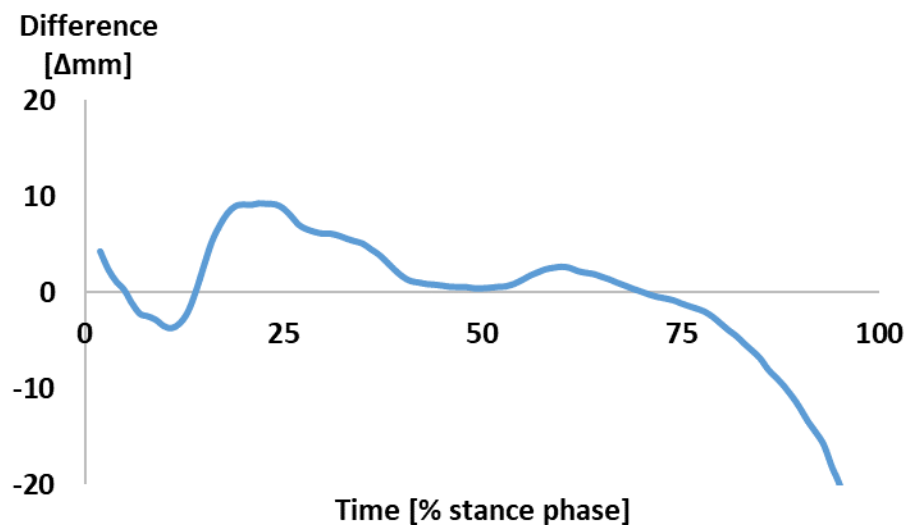


Figure 14 Difference between the mean AP excursion in the Pedar and the Orpyx LogR Gen 2 during a walking trial. A positive value indicates a more posterior COP in the Orpyx LogR Gen 2 insoles, and a negative value corresponds to a more anterior COP in the Orpyx LogR Gen 2 insoles.

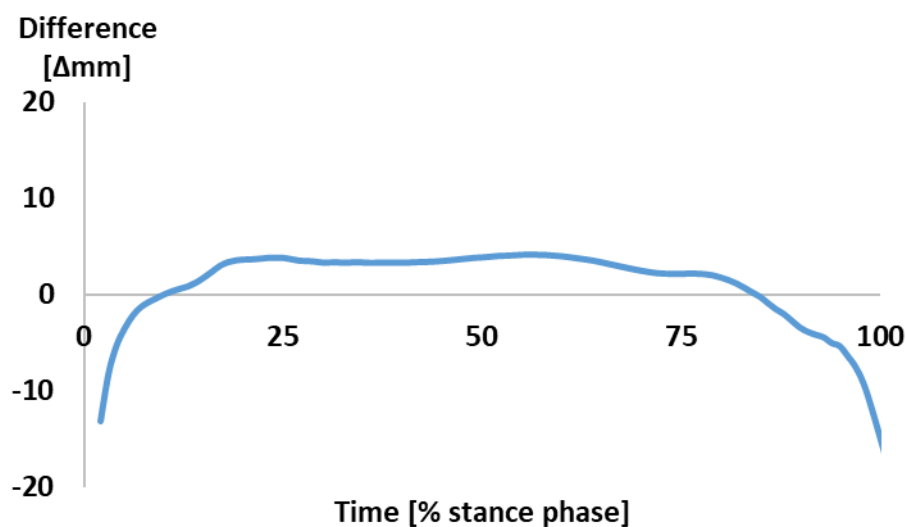


Figure 15 Difference between the mean ML excursion in the Pedar and the Orpyx LogR Gen 2 during a walking trial. A positive value indicates a more lateral COP in the Orpyx LogR Gen 2 insoles, and a negative value corresponds to a more medial COP in the Orpyx LogR Gen 2 insoles.

Side shuffle:

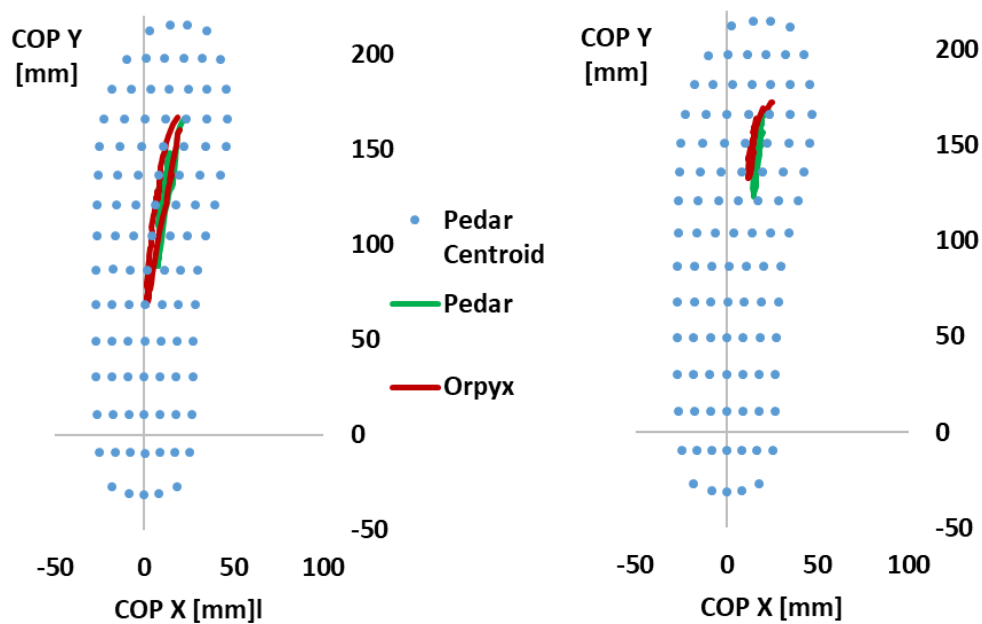


Figure 16 COP trajectories of two different side shuffle movements calculated during two different trials with Pedar (green) and Orpyx LogR Gen 2 (red).

Drop jump:

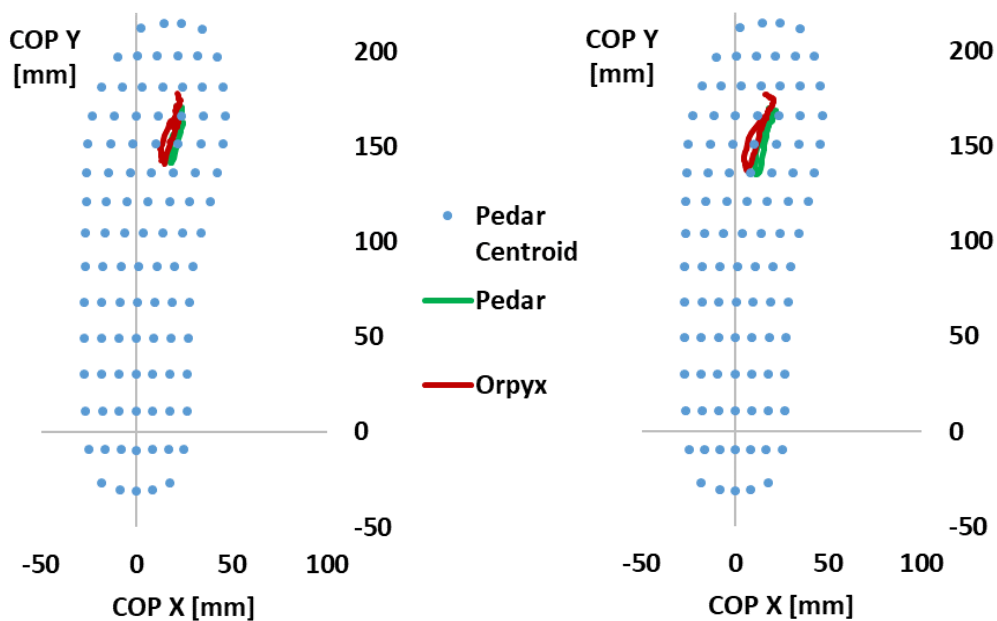


Figure 17 COP trajectories calculated during two different drop jumps with Pedar (green) and Orpyx LogR Gen 2 (red).

Standing:

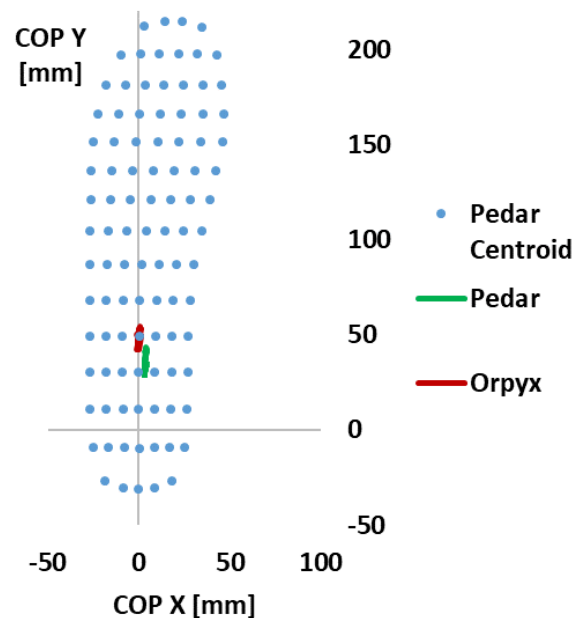


Figure 18 COP trajectories calculated during a standing trial with Pedar (green) and Orpyx LogR Gen 2 (red).

Mean excursion

The results of the mean total excursions, as well as the corresponding standard deviations (SD) and residuals are presented in Table 3 for each task. In Tables 4 and 5, the corresponding mean AP and ML excursions are presented

The results show larger amounts of excursion during the side shuffle, drop jump and standing compared to walking and running.

Table 3 Mean, SD and residual of the total excursion of all steps and trials.

	Mean [mm]	SD [mm]	Residual [mm]
Walking 1	6.16	2.23	554.52
Walking 2	5.23	1.79	471.04
Running	6.41	2.02	576.59
Side shuffle	11.45	11.45	1030.30
Drop jump	13.68	7.15	1231.34
Standing	11.66	11.66	1049.84

Table 4 Mean and SD of the excursion in the AP direction

	Mean [mm]	SD [mm]
Walking 1	4.63	2.43
Walking 2	3.96	2.23
Running 1	5.04	2.29
Side shuffle	10.14	4.71
Drop jump	13.00	7.62
Standing	10.33	3.77

Table 5 Mean and SD of the excursion in the ML direction

	Mean [mm]	SD [mm]
Walking 1	3.21	1.15
Walking 2	2.78	0.93
Running 1	3.08	0.93
Side shuffle	4.23	1.39
Drop jump	2.97	1.33
Standing	4.58	1.68

Step to step variability

The variability between steps during walking in both the AP and the ML directions are plotted as the standard deviation over time (Figures 19-20).

The results show similar amounts of variability in both the AP and the ML directions, with a peak observed at around 40% of the stance phase in the AP direction, and at 55% in the ML direction, respectively. The last 10% of each graph shows large peaks in variability, which can be explained by the small number of cells activated during this portion of the stance phase, resulting in higher sensor errors.

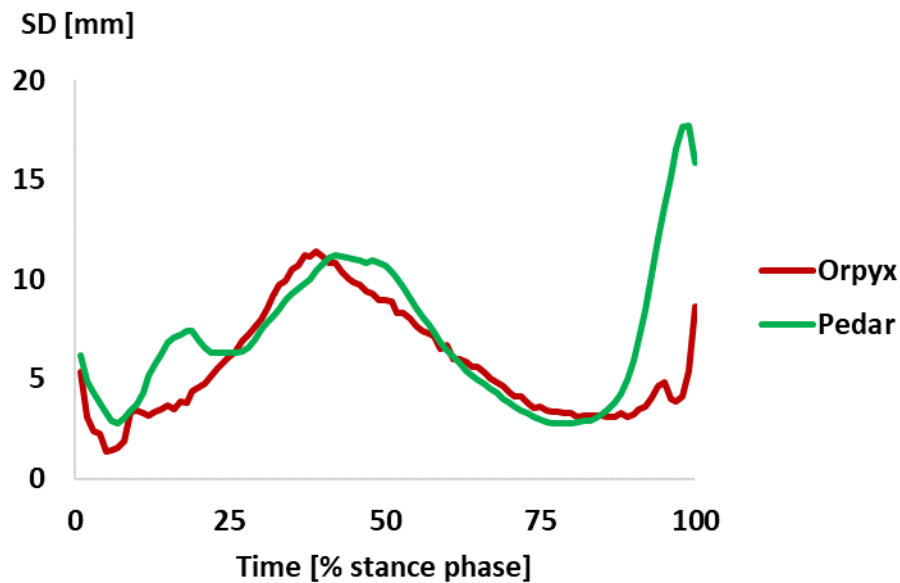


Figure 19 Variability of the COP trajectory in the AP direction during a walking trial calculated from Pedar (green) and Orpyx LogR Gen 2 (red).

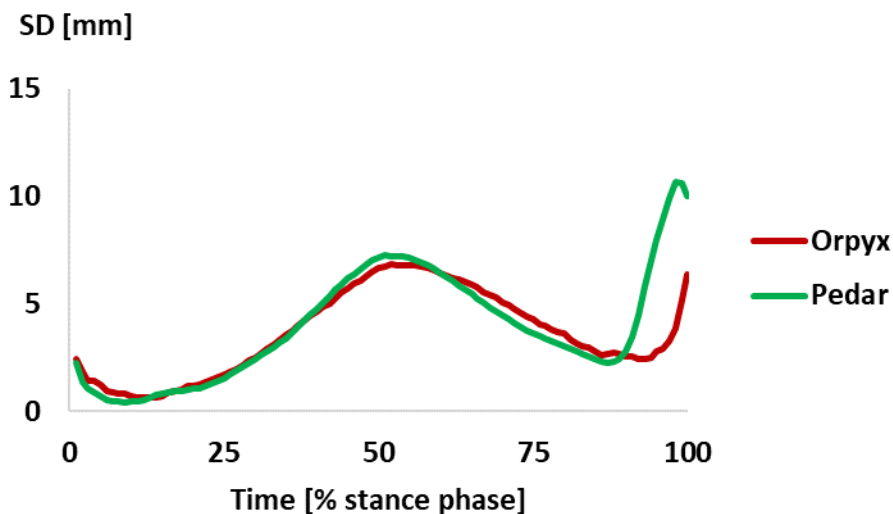


Figure 20 Variability of the COP trajectory in the ML direction during a walking trial calculated from Pedar (green) and Orpyx LogR Gen 2 (red).

Ground reaction force (GRF)

Figures 21 and 22 show the vertical GRF calculated for the mean of 60 (walking) and 96 (running) steps. These results show GRFs about ten times smaller with the Orpyx LogR Gen 2 compared to the Pedar insoles, which was assumed to be resulting from the limited coverage of the sensors of the tested pressure insoles. Therefore, data was estimated for the uncovered area of the Orpyx LogR Gen 2, which did not

change the results significantly. This large difference in GRF may thus result from a calibration error in the Orpyx LogR Gen 2 prototypes used during the corresponding testing sessions.

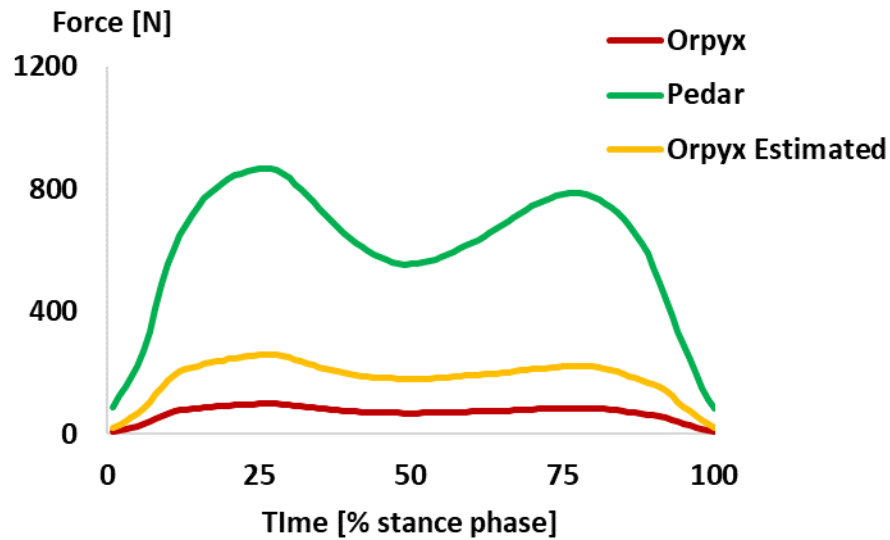


Figure 21 Vertical GRF for the mean of 60 steps of a walking trial with Pedar (green), Orpyx LogR Gen 2 (red) and Orpyx LogR Gen 2 with estimated values for the uncovered areas (orange).

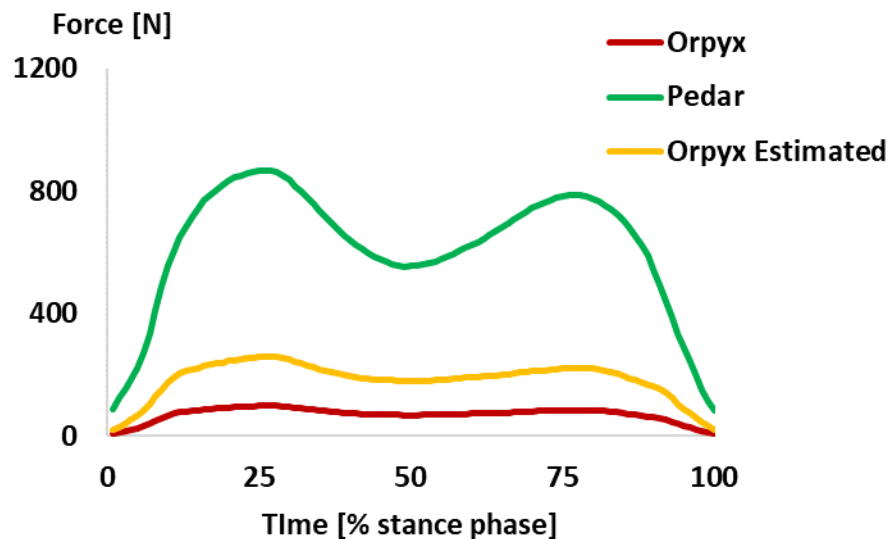


Figure 22 Vertical GRF for the mean of 96 steps of a running trial with Pedar (green), Orpyx LogR Gen 2 (red) and Orpyx LogR Gen 2 with estimated values for the uncovered areas (orange).

Limitations

Due to differences in insoles shapes, the position of the sensors could not be perfectly aligned. This might explain the lateral displacement of the Orpyx LogR Gen 2 insole compared to the Pedar insole.

In addition, it was not always possible to identify the exact same time point for ground contact due to different and low sampling rates. This might lead to an anterior or posterior shift of the COP of the Orpyx LogR Gen 2 compared to the Pedar System.

Study 4- Comparison of the Orpyx LogR Gen 2 and Pedar system to a Kistler force plate

The RMSE and the difference in peak forces calculated between both pressure insole systems and the force plate are presented in Table 6. The difference in RMSE between the Orpyx and the force plate was ~40N larger than the same difference reported for the Pedar compared to the force plate. Similarly, the peak force was ~10% higher in the Orpyx-force plate comparison than in the Pedar-force plate comparison.

Table 6 Mean and SD of the RMSE and difference between peak forces between two pressure insole systems and a standardized force plate.

	Orpyx vs force plate		Pedar vs force plate	
	RMSE [N]	Difference between peak forces [%]	RMSE [N]	Difference between peak forces [%]
Mean	176.9	28.8	135.8	18.6
SD	25.0	5.6	8.3	2.2

The vertical GRF obtained from each system is presented in Figures 23 and 24 during a single walking step. The shape of the GRF profile is similar between systems, but smaller for the pressure insole devices, especially for the Orpyx LogR Gen 2 insoles.

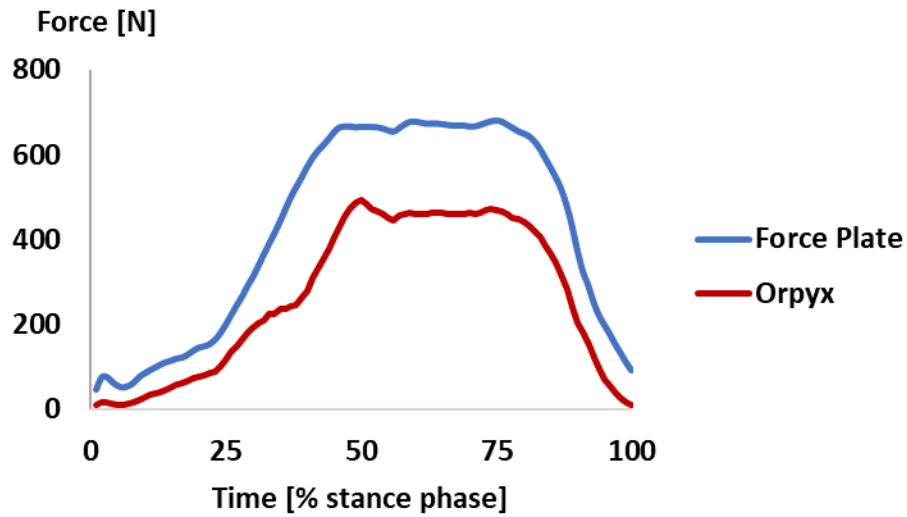


Figure 23 Vertical GRF during a single walking step calculated using Orpyx LogR Gen 2 (red) and a Kistler force plate (blue).

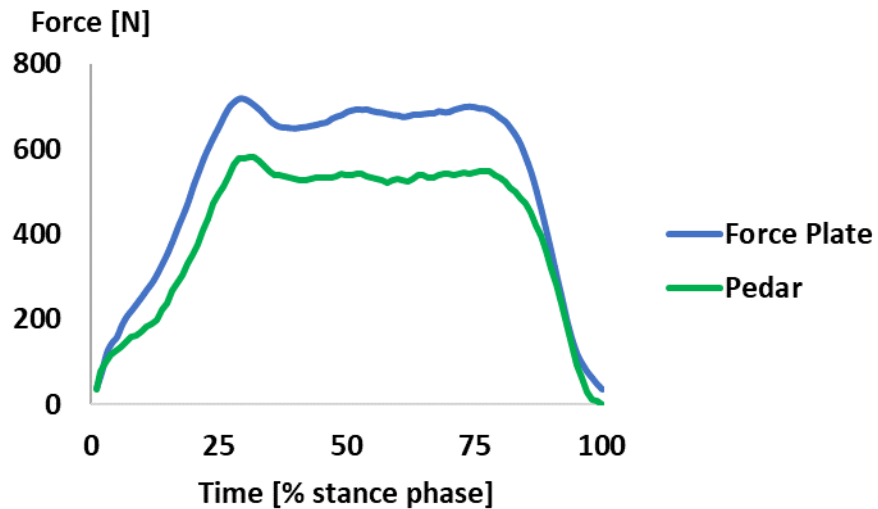


Figure 24 Vertical GRF during a single walking step calculated using Pedar (green) and a Kistler force plate (blue).

Discussion

Four pilot studies were conducted in order to compare the newly introduced Orpyx LogR Gen 2 with a leading competitor (i.e. the Pedar system). In study one, different sensors configurations were artificially tested using Pedar data and the resulting COP trajectories were compared. The results showed a high level of similarity when comparing the original data from the 99-sensor configuration to the interpolated data using artificially generated sensors. This indicates that the difference in number of sensors between Pedar and the Orpyx LogR Gen 2 should not impact the resulting COP trajectories significantly.

In study 2, different pressure insoles placements were compared with respect to pressure distribution patterns. The results showed that the variability between steps was much higher than the variability induced by the superposition of the two insoles in the subject's shoes (based on visual assessment). This led to the decision to collect using both insoles simultaneously in the following comparison study.

In study 3, a more in-depth comparison was performed between the Orpyx LogR Gen 2 insoles and the Pedar system. The results showed a slight lateral shift of the COP trajectory of the Orpyx LogR Gen 2. Overall, this study showed promising results in terms of COP trajectory accuracy when comparing both pressure insole systems, especially during running and walking. However, some larger differences were found when looking at the GRF during the stance phase, which may have been the result of a calibration error. A few limitations were also noted during this study. The difference in shape between the two pressure insoles (Orpyx vs. Pedar), and the difference in sensor size and distribution led to technical challenges regarding the alignment of COP trajectories. This may have resulted in an enhanced lateral displacement. In addition, the sampling rates of the two systems were different, which may have resulted in a slight time shift and potential antero-posterior displacement of the COP. It should also be noted that in the literature, most plantar pressure research divide and average data across physiological areas of the foot (e.g. metatarsals, heel, toes, etc.). Therefore, the current distribution of sensors of the Orpyx LogR Gen 2 may be limiting as it does not allow for such a sub-division of the pressure data.

In study 4, the GRF assessed using both pressure insole systems was underestimated when compared to the force plate, especially for the Orpyx LogR Gen 2. However, the different GRF profiles obtained using both pressure insoles were very similar to the profile measured using the standardized force plate (i.e. similar shapes but different amplitudes).

In summary:

The COP trajectories calculated from artificially generated sensors demonstrated comparable results to the original 99-sensor configuration of the Pedar insoles.

The variability of the pressure distribution patterns between steps and trials was bigger than the variability induced by positioning two pressure insoles on top of each other.

The COP trajectories calculated using the Orpyx LogR Gen 2 insoles were similar to those produced by the Pedar system.

The distribution of sensors in the Orpyx LogR Gen 2 may prevent the division of data into physiologically relevant areas of the foot.

Comparing both insoles to a Kistler force plate showed similar GRF profiles (i.e. shape of the curves) but with lower magnitudes.

Appendix A

Side shuffle:

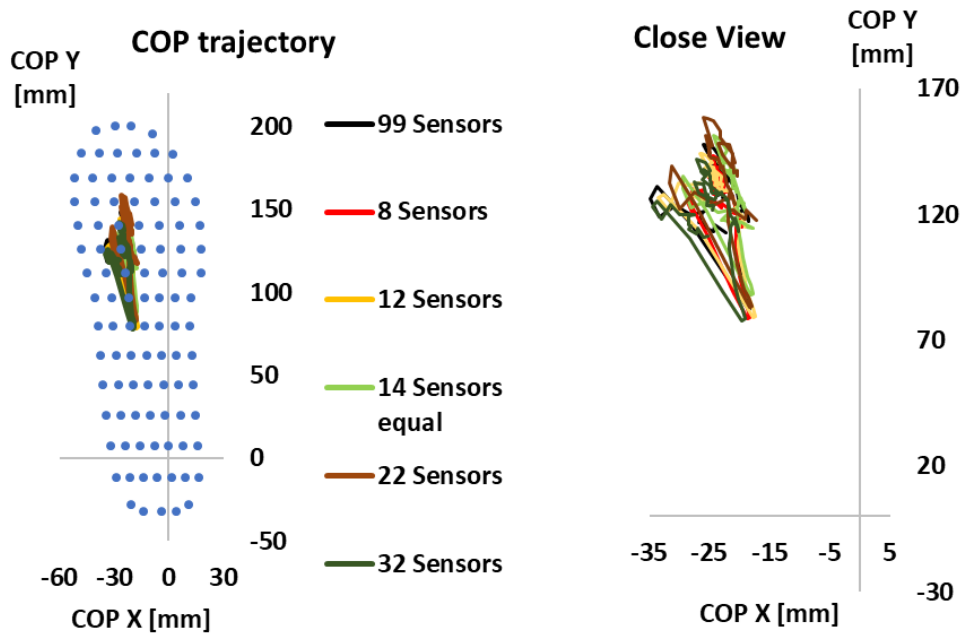


Figure 25 Side shuffle COP trajectories comparing the original 99 sensors to the artificially generated 8, 12, 14 equal, 22 and 32 sensors configurations.

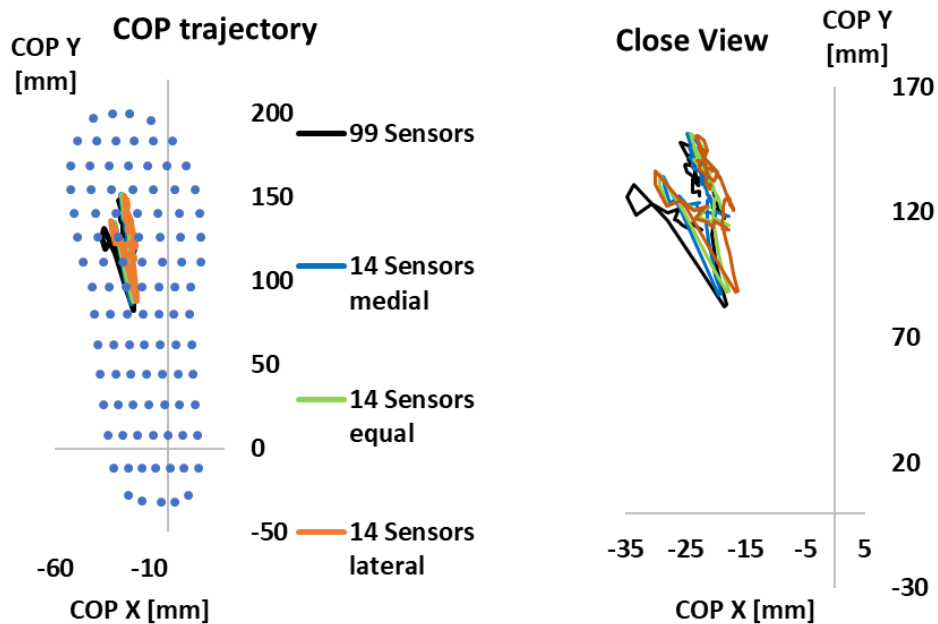


Figure 26 Side shuffle COP trajectories comparing the original 99 sensors to the artificially generated 14 medial, 14 equal and 14 lateral sensors configurations.

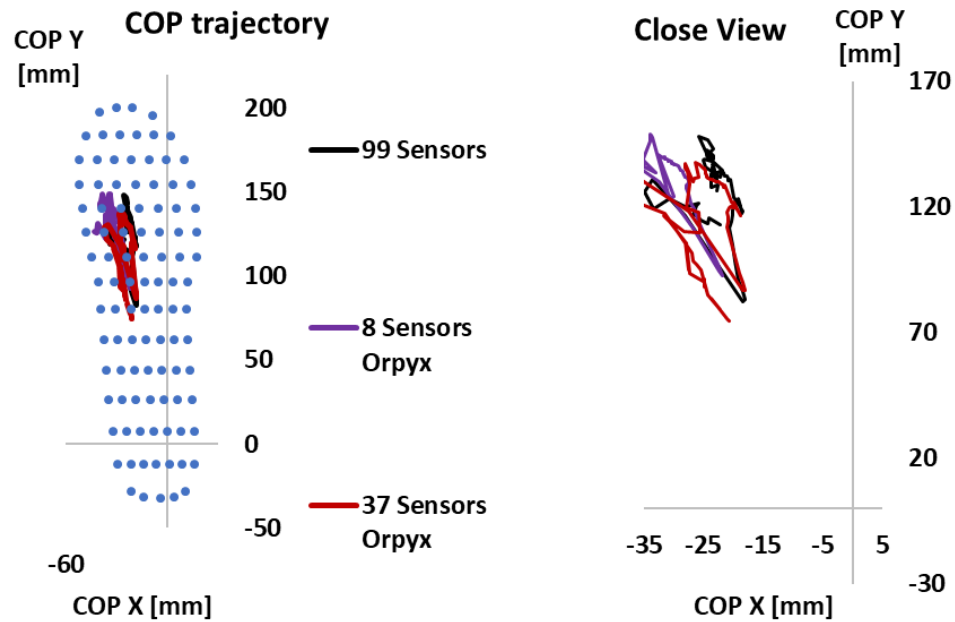


Figure 27 Side shuffle COP trajectories comparing the original 99 sensors to the artificially generated 8 and 37 sensors configurations replicating the distribution of sensors from the two Orpyx insoles designs.

Drop jump:

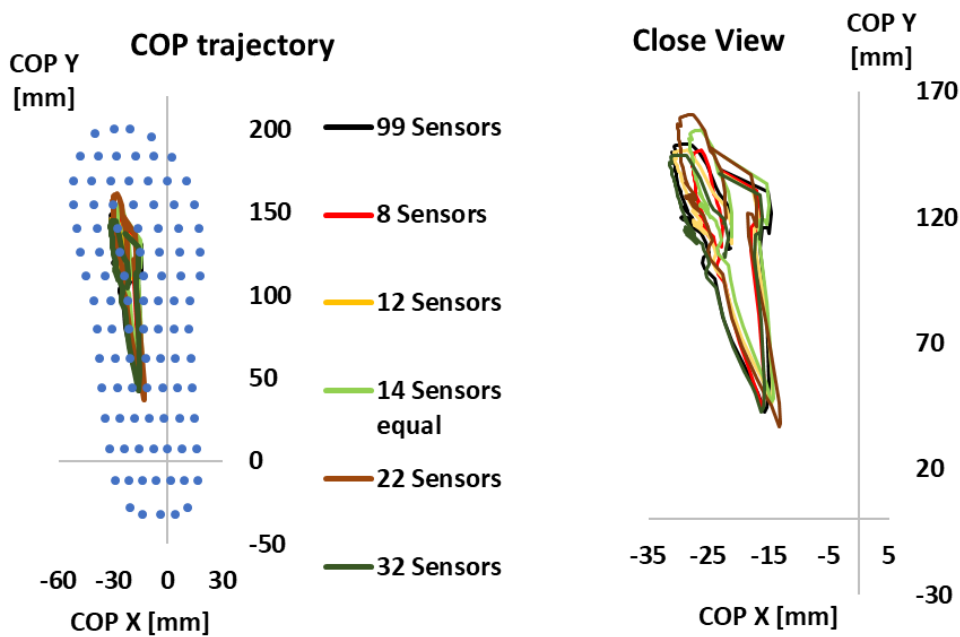


Figure 28 Drop jump COP trajectories comparing the original 99 sensors to the artificially generated 8, 12, 14 equal, 22 and 32 sensors configurations.

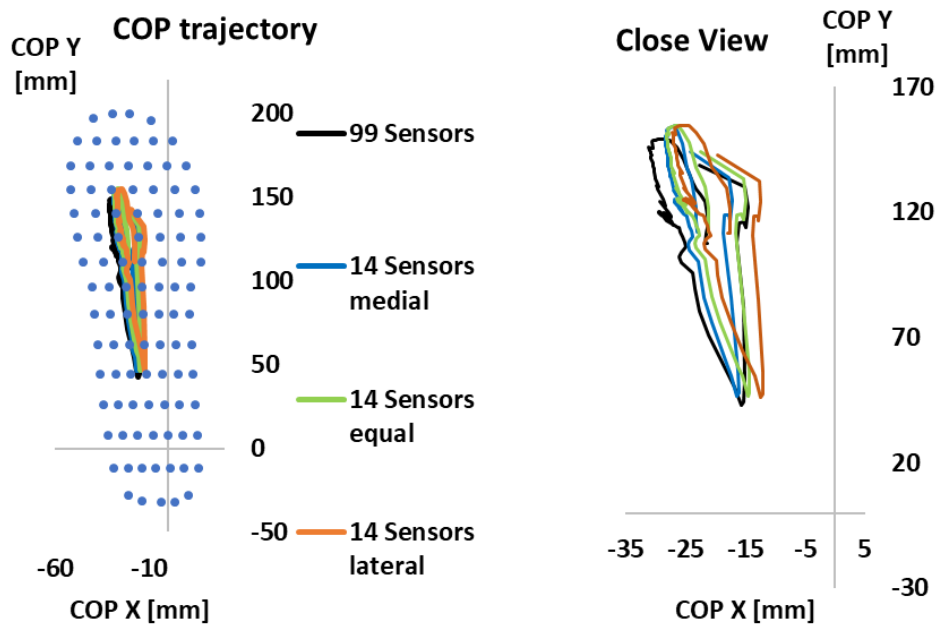


Figure 29 Drop jump COP trajectories comparing the original 99 sensors and to the artificially generated 14 medial, 14 equal and 14 lateral sensors configurations.

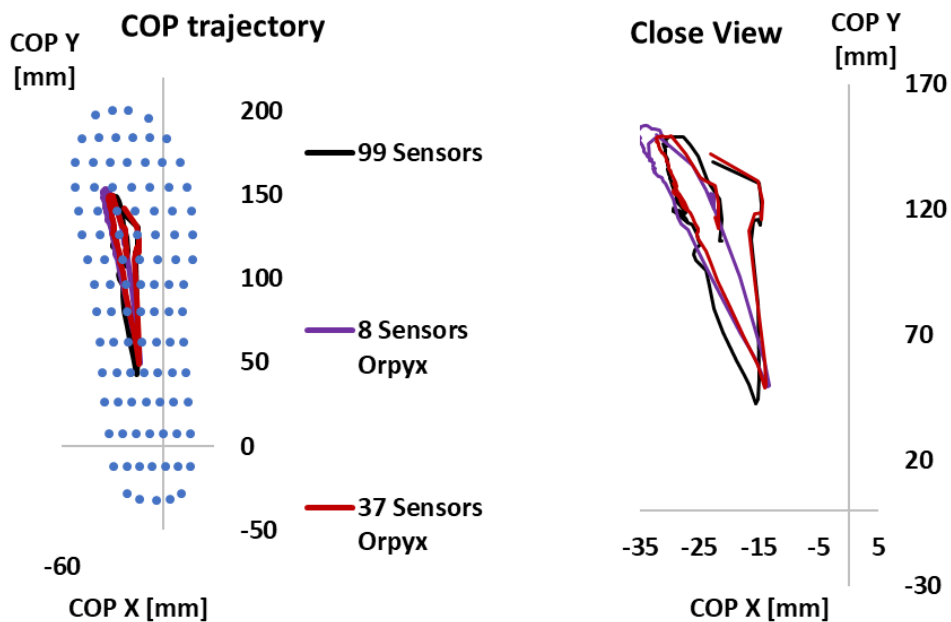


Figure 30 Drop jump COP trajectories comparing the original 99 sensors to the artificially generated 8 and 37 sensors configurations replicating the distribution of sensors from the two Orpyx insoles designs.

Standing:

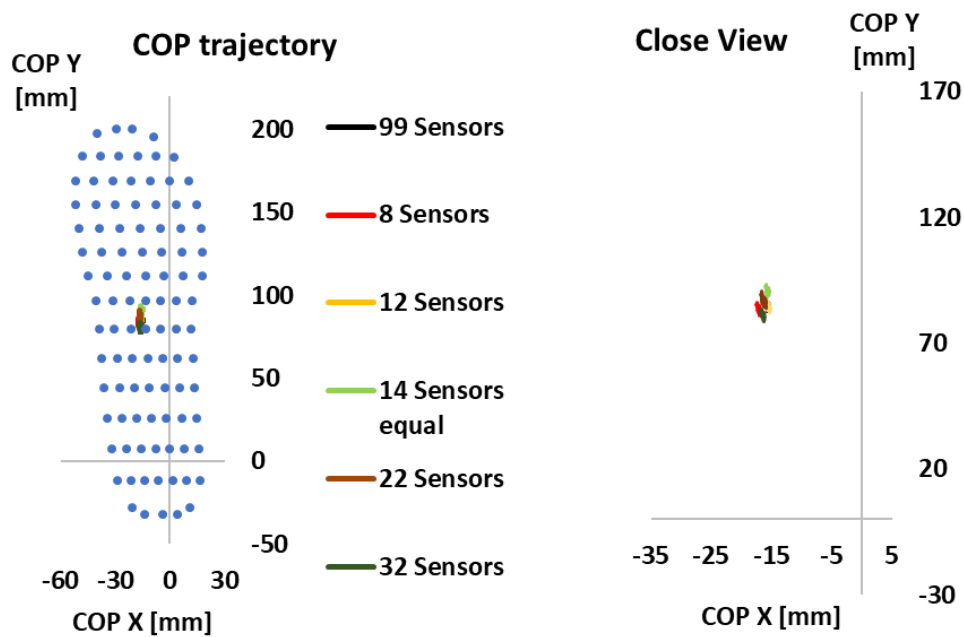


Figure 31 COP trajectories calculated during a standing trial comparing the original 99 sensors to the artificially generated 8, 12, 14 equal, 22 and 32 sensors configurations.

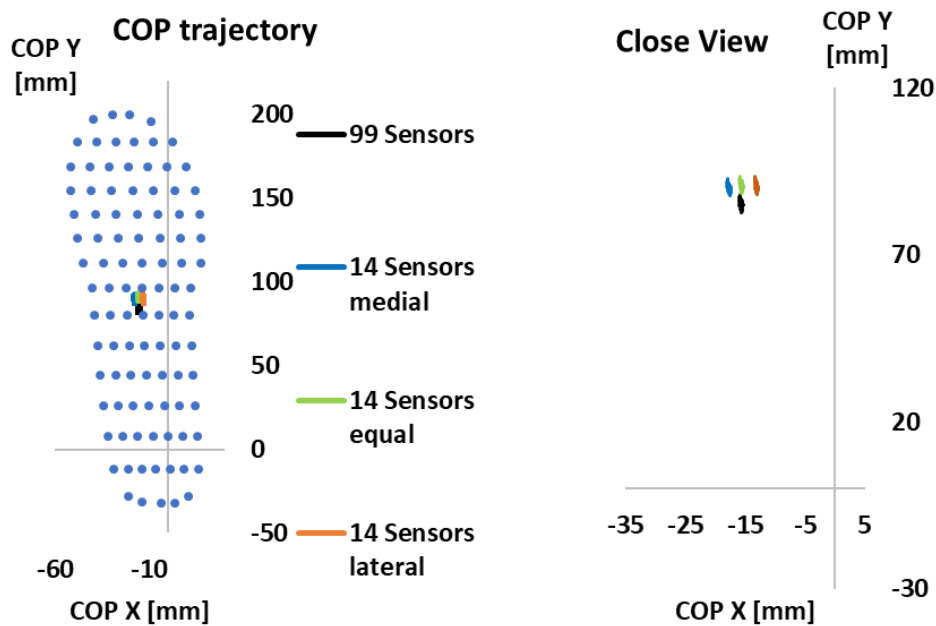


Figure 32 COP trajectories calculated during a standing trial comparing the original 99 sensors and to the artificially generated 14 medial, 14 equal and 14 lateral sensors configurations.

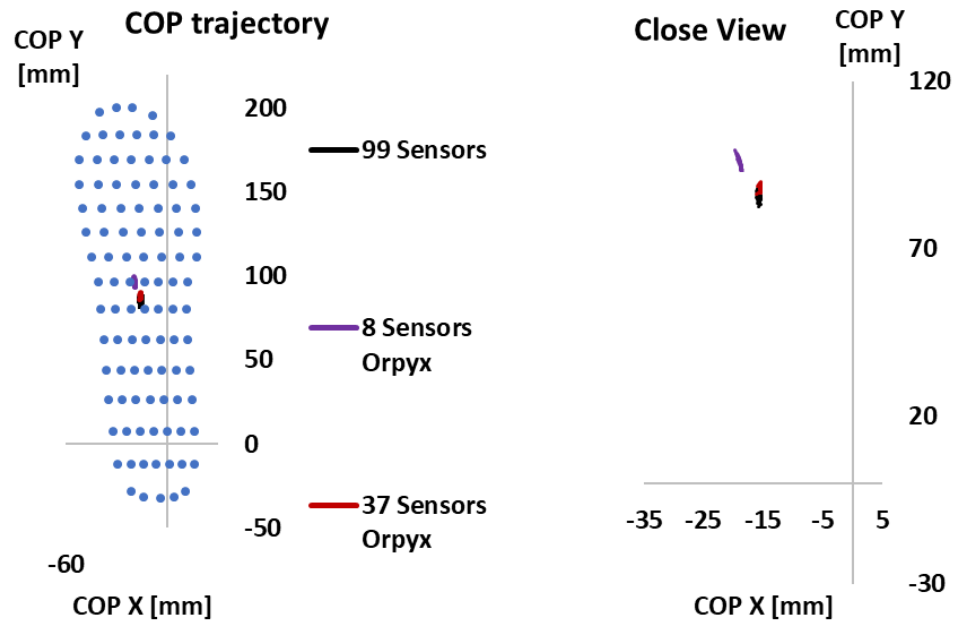


Figure 33 COP trajectories calculated during a standing trial comparing the original 99 sensors to the artificially generated 8 and 37 sensors configurations replicating the distribution of sensors from the two Orpyx insoles designs.